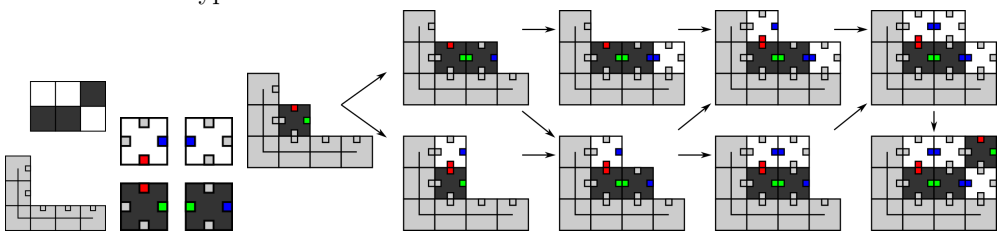
Pattern Assembly Tile Problems

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Figure 1 shows an example Pattern Assembly Tile System (PATS) to reference. In a PATS system, the goal is to build a $w \times h$ pattern of colored tiles using simple tiles and an "L"-shaped seed. Figure 1a shows the input for a 3×2 pattern, and the L-seed to build the pattern.

Every tile has 4 glues- one on each side, and we let each tile have a base color. There are an infinite number of each tile typle. Figure 1b shows a tile set to build the pattern. Every tile must match two glues to stick to the assembly, and thus, they all start with a single tile. Figure 1c shows the possible assembly sequences. Labels, rather than colors, can be used to signify glues. This example shows that the pattern can be built with four tile types. Is this minimal for this pattern? Is there a 3×2 2-color pattern that requires six different tile types.



(a) Input **(b)** Tiles **(c)** Example Assembly Sequence

PATS Size	# of Colors	Number of Tile Types
1×1	1	1
1×2	2	2
2×2	2	?
2×3	2	?

(d) PATS Sizes

Figure 1: (a) Examply Tile Automata system. (b) Smallest sets of states for making $1 \times n$ lines for $1 \leq n \leq 4$. For all of these, the seed state is A. (c) The assembly sequence. (d) Known results for small PATS rectangles with only k colors. Note that the patterns are symmetric and $i \times j$ will have the same tile set result as $j \times i$.

Problems. In [3], they proved that knowing whether a binary pattern can be built with only k different tile types is NP-hard. This was done by utilizing a specific 13-tile pattern that they proved requires 13 unique tiles, which was shown through brute-force computation to guarantee there were no smaller solutions.

In a similar vein, there is no list of "hard" patterns that require unique tiles for each spot in the pattern. The goal of this is to find and catalogue patterns of each size, for small $w \times h$ patterns, that require more tile types (or equal) to any other pattern of that size. Table 1d shows an idea of the results would look. For simplicity, since the complexity is the same, focusing on just two colors ($k = 2$) is probably best.

This is a combinatorial problem if we just brute force all possible systems. Assume the 'L' seed has the same glue at every position. Assuming the desired pattern is a $w \times h$ pattern. Then the number of possible deterministic systems (where glue labels must match to bond) has the following combinations.

- wh possible glues. There are $4wh$ different glues, but assuming every pair of adjacent tiles match on a glue, the 4 is divided out.
- $(wh)^4$ possible different tiles based on glues.
- k different colors for each tile.

The total number of systems is $k(wh)^4$, which is possible for small n. If one is found, the system can halt. For small n, most glue combinations are easily discarded since the tiles will not have matching glues.

A version of this program was created before here [2], but with a Java implementation, it took over a week to get a 2×3 result for two colors. Thus, a new version needs to be optimized for speed and removing edge cases quickly, and possibly should take a distributed approach. Although custom software must be built to simulate the pats system, testing the aTAM systems can be done in VersaTile [1].

References

[1] Algorithmic Self-Assembly Research Group (ASARG). Versatile self-assembly simulator, 2017. https://github.com/asarg/VersaTile.

[2] Algorithmic Self-Assembly Research Group (ASARG). Hardest-k-pats-patterns, 2018. https://github.com/asarg/Hardest-k-PATS-Patterns.

[3] Lila Kari, Steffen Kopecki, Pierre-Étienne Meunier, Matthew J. Patitz, and Shinnosuke Seki. Binary pattern tile set synthesis is np-hard. *Algorithmica*, 78(1):1–46, may 2017. arxiv: 1404.0967.